

Overcoming chemotherapy resistance with herbal medicines: past, present and future perspectives

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Abstract Drug resistance is a major problem in cancer chemotherapy and such resistance may be responsible for treatment failure in 90 % of patients with metastatic cancer. From the research work in the past 30 years, multi-mechanisms responsible for the development of drug resistance have been identified. However, to date single agents that target specific single mechanisms of resistance have not been proven effective. Theoretically, herbs have the potential to target multi-mechanisms of resistance since they contain multiple components and may provide an exciting potential in overcoming drug resistance. The present paper provides an overview of the known mechanisms of resistance and reviews the existing data on herbal medicines (Chinese medicines) as chemosensitizing agents from both the English and Chinese literature. Our review found that certain herbs are capable of inducing strong chemosensitizing effect with various mechanisms, but relevant information useful for development of herbs as viable products for

therapeutic use is generally inadequate. Ideas for improving in vitro screening and animal/clinical studies that could enhance future development of herbal product as chemosensitizing agent for the treatment of resistant cancer are also discussed.

Keywords Chemosensitizing agent · Chinese medicine · Chemotherapy · Drug resistance · Herbal medicine

Introduction

Drug resistance is one of major problems in cancer chemotherapy. Resistance can occur intrinsically (innate resistance) or acquired following a treatment which is initially effective. It is believed that drug resistance is responsible for treatment failure in over 90 % of patients with metastatic cancer (Longley and Johnston 2005). If drug resistance can be overcome, the impact on cancer survival will be very significant.

One potential approach to overcome drug resistance is to target the mechanisms of resistance. Although it is unclear whether the mechanisms of intrinsic resistance are different from that occurred during acquired resistance, similar mechanisms of resistance are likely to occur with both and can be targeted. The general mechanisms of resistance are now well understood (based on the past 30 years of investigation/research) and can include the following:

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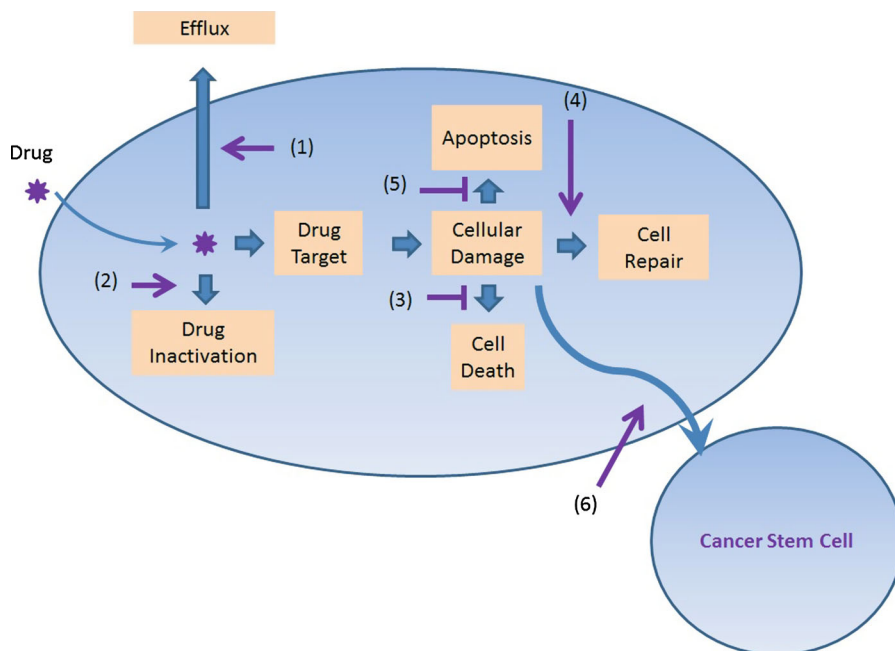


Fig. 1 Overview of mechanisms for developing multidrug resistance to cancer chemotherapy. (1) over-expression of resistant transporter that can enhance drug efflux; (2) inactivation of chemotherapeutic drug such as by high level of thiol glutathione; (3) activation of cell survival signal and inhibition of cell death, such as epidermal growth factor receptor (EGFR);

(4) activation of cellular repair system, such as increase expression of excision repair cross-complementing 1 protein (5) anti-apoptosis such as via NFκB activation; (6) development of resistant cancer stem cells leading to both innate and acquired drug resistance. (Adapted from Longley et al. 2005)

increased drug efflux and decreased drug influx, drug inactivation, alterations in drug target, processing of drug-induced damage, and evasion of apoptosis (Longley and Johnston 2005). Some examples of specific mechanisms are: (1) expression of resistant transporters (e.g. MDR1/Pgp, BCRP, MRPs) or genes that can enhance drug efflux (Huang et al. 2005; Makarovskiy et al. 2002; Wang et al. 2010b); (2) inactivation of chemotherapeutic agents, e.g., high level of thiol glutathione that can inactivate platinum drugs, i.e. via NF-E2-related factor 2, Nrf2, a redox-sensing transcription factor (Singh et al. 2010; Wang et al. 2010a); (3) activation of cell survival signal and inhibition of cell death, e.g., overexpression of the oncoprotein epidermal growth factor receptor (EGFR) (Salzberg et al. 2007; Sirotinak et al. 2000); (4) activation of cellular repair system, e.g., increase expression of excision repair cross-complementing 1 protein (Longley and Johnston 2005) (5) anti-apoptosis, e.g., NFκB activation exerting anti-apoptotic effect (Longley and Johnston 2005) and (6) development of resistant cancer stem cells (CSC) leading to

both innate and acquired drug resistance via changes in CSC-specific pathways such as Notch, Wnt, and Hedgehog pathways (see Fig. 1) (Crea et al. 2011; Domingo-Domenech et al. 2012).

The existence of these multi-mechanisms may explain the observation that single molecules which target single mechanisms of resistance have not been proven effective in overcoming chemotherapy resistance as of to date (Baumert and Hilgeroth 2009; Robey et al. 2010; Yuan et al. 2008).

Herbal medicines contain multiple components which could target multiple mechanisms of resistance. Thus, herbal medicines may have greater potential to successfully overcome chemoresistance than single agents. This article will first define the chemosensitizing effect and then review chemosensitizing herbs (or Chinese medicines and related components) reported from in vitro, in vivo, and clinical studies in English and Chinese literature. A large database PubMed from 1960 to 2013 with key words of “herb and MDR” as well as a well-known Chinese data base (China Journal Net) from 1980 to 2013 were utilized to

perform the literature search. In view of the exciting potential of developing chemosensitizing herbs as viable products for therapeutic use, considerations for improving in vitro screening and animal/clinical studies that could enhance successful product development for future treatment of resistant cancer are also discussed.

Definition of chemosensitizing effect (CE)

CE can be defined as the ability of the chemical substance(s) to restore or improve the anticancer effect in cells that are resistant to the initial chemotherapy (see equation below). Thus the CE is a property of the herb that specifically re-sensitizes a drug effect when used in combination. In general, any herb-drug combination may lead to synergistic, additive, or antagonistic effect. A synergistic effect occurs when the combination generates an overall effect that is greater than the sum of the individual effect. Thus the CE is a special type of synergistic effect: for the CE induced by the herb in a given herb-drug combination, the two agents may be independent mechanistically and the herb itself may have no anticancer (i.e. cytotoxic) effect. This condition appears to fit into the Chinese medicinal holistic theory, with the chemotherapeutic agent acting as “Sovereign” and the herb as “Minister” (Che et al. 2013).

The potency of effect from an herb-drug combination has been commonly assessed by the use of the combination index (CI) (Chou 2006). The determination of CI usually assumes each agent to exert some common activity (e.g. cytotoxic activity). Thus, instead of using CI, an operational definition of CE reflecting the effects from the herb-drug combination is more desirable since the herb may have no cytotoxic effect of its own at the given concentration used. The operational definition of CE (based on in vitro experiments using cancer cell lines) can simply be the measure of the degree to which an herb or substance that can lower (i.e., chemosensitize) the concentration of a chemotherapeutic drug's half maximal inhibitory concentration (IC_{50}) when the herb and the chemotherapeutic drug are co-administered. (IC_{50} is the concentration of a chemotherapeutic drug that inhibits cell proliferation by 50 %). Quantitatively, the CE can be expressed as: $CE = IC_{50D}/IC_{50H+D}$, where IC_{50D} is the IC_{50} of the chemotherapeutic drug against which

an herbal extract (or other substance suspected of having a CE) is evaluated; while IC_{50H+D} is the IC_{50} of the combination of the herb and chemotherapeutic drug. The CE can vary depending on the type of cell, the specific chemotherapeutic agent or the combination that is being evaluated. A similar concept has been also used by Shen previously (Shen et al. 2005).

Chemosensitizing herbs

The use of herbal medicine to overcome chemoresistance is a relatively new concept. From our search of large English language data base (PubMed) and Chinese literature data base (China Journal Net), about 49 herbal substances (individual Chinese herbs, active components, and complex formula) possessing CE have been reported. Most of these reports on CE are based on in vitro studies. The reported CE of various herbal substances, including individual active components, single herbs (usually with more than one active component) and complex formula (combination of herbs) are summarized in Table 1 and are briefly described below in alphabetical order.

Active herbal components

6-([1,3]dioxolo[4,5-g]isoquinoline-5-carbonyl)-2,3-dimethoxy-benzoic acid methyl ester, 8-oxocoptisine, and palmatine

See Coptis Rhizome *under Single Herbs* below.

Andrographolid Andrographolid (AG) is an active component of *Andrographis paniculate*, which is commonly used in China and East Asian countries for the treatment of infection and fever associated with common cold, tonsillitis and diarrhea. AG also possesses anti-tumor effect (Jayakumar et al. 2013). In a study in HCT-8/5FU cell line (5-fluorouracil (5-FU) resistant subline of colorectal cancer) using 5-FU, adriamycin or cisplatin alone versus combination with AG (1.2 mg/ml) (Han et al. 2005), the inhibition of cell viability was only found to be slightly enhanced by AG. The mechanism of inhibitions was not clear but might be related to Pgp inhibition.

Apigenin, Fisetin, and Honokiol These 3 plant derived sedatives are found in many plants. The CE of apigenin, fisetin, and honokiol was evaluated in

Table 1 Herbal substances and their chemosensitizing effect on specific drugs

Herb/Pure compound	Concentration	Drug	Mechanism and effect	In vitro/in vivo	Reference
<i>Pure compounds</i>					
(6-([1,3]dioxolo[4,5-g]isoquinoline-5-carbonyl)-2,3-dimethoxybenzoic acid methyl ester, 8-oxoptisine, and palmitate)	10 μ M	Paclitaxel	Enhance paclitaxel cytotoxicity	SK-OV-3, HCT15/P-gp, MES-SA/DX5	Min et al. (2006)
Andrographolid (Andrographis paniculate)	1.2 mg/ml	5-FU, Adriamycin, Cisplatin	Inhibition of P-gp activity	HCT-8/5-FU	Han et al. (2005)
Apigenin, fisetin, honokiol	10 μ M	Doxorubicin	Decrease the cellular GSH level and increase the reactive oxygen species; CE 1.0 to 1.7-fold	MES-SA/DX-5	Angelini et al. (2010)
Camptothecin	100 nM	Daunorubicin	CE 13 fold	MCF7/VP (MRP1 overexpressed), KH30 cells	Chauvier et al. (2002)
Cepharanthine (Stephania cepharantha Hayata)	2 μ M	Paclitaxel	Increase of cellular accumulation	HEK293 (MRP7 transfected cell)	Zhou et al. (2009)
Curcumin	5 μ M, 2.5–20 μ M	Adriamycin, Melphalan	Decrease in P-gp expression and increase intracellular drug concentration; CE 1.4-fold	K562/ADM; MOLP-2/R	Xu et al. (2011); Xiao et al. (2010)
Gensenoside Rh2		Daunorubicin and vincristine	The cytotoxicity of daunorubicin and vincristine was enhanced	P388/ADM	Hasegawa et al. (1995)
Ginsenoside Rg3	10, 40 mg/kg	Doxorubicin and Vincristine	Reverse the chemoresistance to doxorubicin and vincristine in KBV20C; increase life span of mice implanted with P388/ADM cells	KBV20C; P388/ADM	Kim et al. (2003)
Luteolin	5 μ M	Cisplatin	Enhance cytotoxicity of cisplatin		Xu et al. (2012)
Myricetin	25 μ M	Vincristine	Modulate MRP1 and MRP2 mediated efflux	MDCKII-MRP1, MDCKII-MRP2	van Zanden et al. (2005)
Parthenolide	20 μ M	Doxorubicin	Inhibition of P-gp mediated efflux; downregulation of P-gp expression via NF-kb pathway; downregulation of HSP70. CE 6.0-fold	A549, A549/Dox	Xin et al. (2013)
Polyphyllin D	0.2–1.4 mM	Cisplatin	Decrease IC50 of cisplatin by 2.1 μ M	OVCA	Al Sawah et al. (2013)
Quercetin	25, 50, 100 and 200 μ M	Cisplatin	CE 1.1 to 3.3 for cisplatin and 1.2 to 3.1 for vincristine	A549/S and A549/VCR	Zhang et al. (2012)
Tetramethylpyrazine	400, 600 μ M	Adriamycin	Down regulation of the expression of P-gp, MRP2 and MPR3; CE 1.6 and 9.2-fold for 2 concentrations	BEL-7402/ADM	Wang et al. (2010a)

Table 1 continued

Herb/Pure compound	Concentration	Drug	Mechanism and effect	In vitro/in vivo	Reference
Theanine	10 mg/kg (theanine)	Doxorubicin	Decrease tumor weight by 68 % compared with doxorubicin alone in mice implanted with P388/Dox	P388/Dox bearing mice	Sadzuka et al. (2000)
<i>Herbal extract</i>					
Coptis Rhizome/Berberine	1.6–51.2 µg/ml (extract); 0.5–16 µg/ml Berberine	5 FU Camptothecin, Paclitaxel	Increase ROS production; Slightly increase cellular drug accumulation. Enhance cytotoxicity of 5 FU, camptothecin, paclitaxel	A549	He et al. (2012)
Fructus Schizandrae (Schizandrin A)	6.25–25 µg/ml (extract); 25–50 µg/ml (Schizandrin A)	Doxorubicin, Vincristine	Inhibition of Pgp activity by down regulating its expression and inhibition of PKC. CE 1,624, 32, and 113 fold for vincristine, doxorubicin and paclitaxel respectively	Bel7402 cells, KB/KBv200, MCF7/MCF7Dox	Huang et al. (2008)
Rhei Rhizoma, Scutellariae Radix, Portia, Zizyphi Fructus, Zingiberis Rhizoma, Asiasari Radix, Sophorae Radix	300 mg/kg	Vincristine	Retard tumor growth in mice implanted with KBv200 cells	Xenograft implanted with KBv200 cells	Takara et al. (2005)
Rosemarinus officinalis Labiatae	10–50 µg/ml	Paclitaxel	Inhibition of Pgp activity; CE 1.7 to 5.0-fold	Hela	
	16.5–82 µg/ml	Doxorubicin, Vinblastine	Increase cellular accumulation of chemotherapeutic agents. Decrease IC50 of vinblastine	R65 (Pgp overexpressed MCF7), MCF clone 10.9 (transfected with MDR1)	Ciolino et al. (1999)
Tripterygium wilfordii	12.5, 25 µg/ml	Daunorubicin, Docetaxel	Downregulation of Pgp level and EGFR; Increase intracellular drug concentration	K562/Dox; PC3-TxR	Wang et al. (2011); Chow et al. (2014)
	500 mg/kg	Docetaxel		Xenograft model with PC3-TxR	
<i>Herbal formula</i>					
Shenghe Powder	10 µg/ml	Vincristine	P-glycoprotein (P-gp), lymphoma/leukemia-2 (Bcl-2),	SGC-7901/vincristine	Wang et al. (2007)
Shen-qi-jian-wei Tang			Inhibit expression of MDR1 and LRP in the peripheral blood of gastric cancer patients	Clinical	Shu et al. (2008)
Si-Wu-Tang		Adriamycin	Increase intracellular drug concentration	K562/ADM	Shen et al. (2005)

Table 1 continued

Herb/Pure compound	Concentration	Drug	Mechanism and effect	In vitro/in vivo	Reference
Supplement Energy and Nourish Lung		Adriamycin	Decrease MRP expression and increase accumulation of intracellular Adriamycin	SW 1573/2R120	Xu et al. (2008)
Yi-qi-yang-ying-fang		Cisplatin	Sensitize the effect of cisplatin to human lung A549/DDP cells	A549/DDP	Sun et al. (2009)
Zuo Jin Wan (Rhizoma Coptidis and Fructus Evodiae)	25–100 µg/ml; 1,027.5– 4,110 mg/kg	Oxaliplatin,	Diamminedichloroplatinum, Mitomycin, 5Fu	Downregulation of Pgp; CE 2.2, 1.8, and 1.4-fold for oxaliplatin, diamminedichloroplatinum, and mitomycin respectively.	HCT116/L-OHP; Xenograft

Sui et al. (2013)

human doxorubicin resistant uterine sarcoma cells (MES-SA/Dx5). These compounds (10 µM each) were found to enhance doxorubicin activity 1.7, 1.2, 1.4-fold, and 1.2, 1.0, 1.1-fold when combined with doxorubicin 2 µM and 8 µM respectively. These three compounds also increased the intracellular doxorubicin concentration. Apigenin in addition decreased the cellular GSH level and increased the reactive oxygen species. The results indicated that these compounds could be potential active chemosensitizing agents (Angelini et al. 2010).

Berberine

See Coptis Rhizome under *Single Herbs below*.

Camptothecin Camptothecin, a DNA enzyme topoisomerase I inhibitor, is an active component from the bark and stem of *Camptotheca acuminata* (Camptotheca, Happy tree) and has been used as an effective anticancer drug. Although camptothecin is a cytotoxic agent, the combination of camptothecin and daunorubicin has also been reported to overcome multidrug-resistance in MRP1 overexpressed breast cancer MCF-7 cells (MCF7/VP) and leukemia (KH30 cells), producing enhanced cytotoxicity and nuclear accumulation of daunorubicin (Chauvier et al. 2002).

Cepharanthine Cepharanthine (CEP) is a natural alkaloid isolated from *Stephania cepharantha* Hayata with antibacterial activity. The combination of CEP and Paclitaxel was investigated in MRP7-transfected HEK293 cell line. The addition of cepharanthine 2 µM to paclitaxel was found to induce CE 13 fold (described as dose modifying factor in the study) compared to paclitaxel alone. In the HEK293 parent cells, cepharanthine was found to also decrease the IC₅₀ of paclitaxel. Cepharanthine did not alter the expression of MRP7 but increased the cellular accumulation via competitive inhibition (Zhou et al. 2009).

Curcumin Curcumin is a phenolic coloring compound, extracted from the rhizomes of *Curcuma longa* Linn. This compound is known to induce apoptosis of tumor cells with exhibition of suppression, retardation, or inversion of carcinogenesis (Xiao et al. 2010). In a study in K562 cells resistant to adriamycin, a low concentration of curcumin (5 µM) + adriamycin was able to induce CE 1.4-fold compared to adriamycin alone (Xu et al. 2011). The mechanism of action was

found to be related to a decrease in Pgp expression based on the observation of decreased MDR1 mRNA levels as well as increase in intracellular rhodamine123 accumulation (a Pgp substrate) observed in the resistant K562 cells (Xu et al. 2011) and resistant cervical carcinoma cell line (KB-V1) (Anuchapreeda et al. 2002).

The combination effect of curcumin and melphalan was also studied in the multidrug resistant human multiple myeloma cell line (MOLP-2/R) (Xiao et al. 2010). The combination was found to be synergistic in the MOLP-2/R cells (but not the sensitive cells) at curcumin concentrations 2.5 to 20 μM . A CE of 2.4 was calculated in the presence of 10 μM curcumin. The IC_{50} of curcumin alone was around 8 and 15 μM in MOLP-2 and MOLP-2/R cells respectively. The mechanism was found to be related to the inhibition of FA/BRCA pathway by down regulation of FANCD2 protein monoubiquitination (Xiao et al. 2010).

Ginsenoside Rg3 and Gensenoside Rh2 These are well known components of *Radix ginseng*. Ginsenoside Rg3 was found to be capable of reversing the resistance to doxorubicin and vincristine in vincristine-resistance KBV20C cells, most likely by modulating drug accumulation and efflux (Kim et al. 2003). In vivo, the combination of ginsenoside Rg3 (10 mg/kg and 40 mg/kg) with docetaxel (4 mg/kg) was found to significantly increased the life span of BDF1 mice implanted with docetaxel-resistant murine leukemia P388 cells (Kim et al. 2003). Also ginsenoside Rh2 was found to enhance the cytotoxicity of daunomycin and vinblastine in docetaxel-resistant murine leukemia P388 cells (P388/ADM) (Hasegawa et al. 1995).

Luteolin, quercetin Luteolin and quercetin are two common flavones found in various fruits, vegetables, leaves and grains. Luteolin possesses multiple activities and can be utilized as an antioxidant, a free radical scavenger, a promoter of carbohydrate metabolism, and so on. Quercetin also has similar property and has been used as an ingredient in supplements, beverages, or foods. In a recent study, luteolin was found to enhance the cytotoxicity of cisplatin in A549 cells at concentration of 5 μM (Xu et al. 2012). This chemosensitizing concentration was 6 fold lower than its IC_{50} in A549 cells (Xu et al. 2012). Also in recent study, quercetin was found to increase the chemosensitivity of A549/S and A549/VCR cell lines to cisplatin and vincristine at concentrations 25, 50,

100 and 200 μM (with CE values of 1.1, 2.1, 3.3, and 3.9 for cisplatin, and 1.2, 1.7, 2.3, and 3.1 for vincristine respectively (Zhan et al. 2012).

Myricetin Myricetin is a natural flavonol found in many grapes, berries, fruits, vegetables, herbs, as well as other plants. It has been reported to reverse in vitro cellular MRP1 and MRP2 mediated vincristine resistance (van Zanden et al. 2005).

Parthenolide Parthenolide, a sesquiterpene lactone, is an active component found in *Tanacetum parthenium* or feverfew, which is widely used for inflammatory diseases, migraine, menstrual irregularities, and fever. Parthenolide has been reported to sensitize the resistant human lung adenocarcinoma epithelial cell line A549 (A549/Dox) to doxorubicin with a CE value of 6.0 obtained from parthenolide + doxorubicin combination. The potential mechanisms appeared to be related to (1) the reduction of MDR1 transporter (Pgp) expression via inhibiting NF- κB signaling pathway as evidenced by the decrease of rhodamine 123 (a substrate of Pgp) efflux in A549/Dox cells, and (2) reduction of the HSP70 (heat shock protein). (Xin et al. 2013) (Heat shock protein is a well-known chaperone family with cytoprotective effects involved in the development of MDR).

Polyphyllin D Polyphyllin D is an active component of *Paris polyphylla* and is known to be a potent cytotoxic agent. Polyphyllin D was found to significantly decrease cisplatin IC_{50} in OVCA cells (mean IC_{50} reduction of 2.1 μM) (Al Sawah et al. 2013).

Schizandrin A and 4 other active components of Fructus Schizandrae

See *Fructus schizandrae* under *Single Herbs* below.

Tetramethylpyrazine Tetramethylpyrazine (TMP) is an active component of *Ligusticum chuanxiong* Hort, a Chinese herb (Chuanxiong) (Wang et al. 2004). However in another report, no such compound has been detected (Wang et al. 2009). Chuanxiong Rhizoma, the dried rhizome of *Ligusticum chuanxiong* Hort, has been widely used for promoting blood circulation, alleviating pain, as well as for cardiovascular diseases (Li et al. 2012). In a recent study, TMP at concentrations of 400 and 600 μM was found to inhibit BEL-7402/ADM cell (adriamycin resistant cell line) proliferation with CE of 1.6 and 9.2-fold

respectively when used in combination with adriamycin (Wang et al. 2010a). The underlying mechanism was most likely related to the down regulation of the expression of Pgp, MRP2 and MRP3 leading to increased accumulation of adriamycin (Wang et al. 2010a). In this study, the cell line resistant index (the ratio of IC₅₀ of resistant cells over the sensitive cells) was only 1.53, so the observed CE is unlikely to be very significant.

Theanine

See Green Tea under *Single Herbs below*.

Polysaccharides Some polysaccharides from Chinese medicine have been also found to possess CE. Astragalus polysaccharides, an active component of Radix Astragali with no anti-tumor activity itself, was found to increase the cytotoxicity of cyclophosphamide, adriamycin, 5-Fluorouracil, cisplatin, etoposide, and vincristine in H22 hepatoma cells resistant to adriamycin. The observed effect was most likely related to downregulation of MDR1 mRNA expression and inhibition of Pgp efflux pump function (Tian et al. 2012). Also, certain polysaccharide from Lingzhi (a well know immunomodulating herb) was observed to profoundly reverse cisplatin resistance and arsenic resistance in urotheliu carcinoma cells. The mechanism was probably related to down-regulation of Akt and activation of p38, as determine by MTT assay and median-effect analysis (Huang et al. 2010).

Single herbs

Coptis rhizome

The rhizome of *Coptis chinensis* Franch. (named as *Rhizoma Coptidis* in the Chinese Pharmacopeia, with the Chinese name Huang Lian) and its active component berberine (BER) have been widely used in various formulas against intestinal infection, diarrhea and inflammation. In a recent study, combinations of this herb or BER with chemotherapeutic agents including 5-FU, camptothecin, or paclitaxel were found to inhibit A549 cell growth more effectively compared to chemotherapeutic agents alone (He et al. 2012). From these study results, the herb camptothecin showed a synergistic effect but not with 5-FU or

paclitaxel. The mechanism of the observed CE was postulated to be related to the increase in reactive oxygen species (a known mechanism that sensitize cancer cells to drugs). Although the herb or BER also increased the intracellular concentration of rhodamine 123, a Pgp substrate, the extent was small and thus would unlikely be the mechanism responsible for the CE (He et al. 2012).

Three other pure components from *Coptis Rhizome* (*Coptis japonica* Makino in Japan): (6-([1,3]dioxolo[4,5-g]isoquinoline-5-carbonyl)-2,3-dimethoxy-benzoic acid methyl ester, 8-oxocoptisine, and palmatine were also found to sensitize paclitaxel cytotoxicity in SK-OV-3 (human ovarian cancer), HCT15 (human colorectal cancer with overexpressed P-gp), and MES-SA/DX5 (human MDR uterine sarcoma) cells. (Min et al. 2006). More studies are needed to confirm these results as these three compounds only showed marginal or non cytotoxic effect with no CE observed from the paclitaxel combination.

Fructus Schizandrae

Fructus Schizandrae (FS) is commonly used as a tonic. Recently, FS has been found to significantly improve liver dysfunction in patients with chronic hepatitis. The chemosensitizing effect of crude extract of FS (LCC) and its five main active components have been investigated using KB cells (human epidermal carcinoma) and its MDR subline (KBv200), MCF7 and its MDR subline (MCF7/Dox), as well as Bel7402 (human hepatic cellular carcinoma) cell line. The LCC was found to significantly sensitize the MDR cell lines to chemotherapeutic agents vinblastine and doxorubicin not only in the resistant cell lines but also in the sensitive cells in a dose dependent manner (in the concentration range from 6.25 to 25 µg/ml). The CE (or fold reversal as reported in the article) of LCC in combination with vincristine ranged 8–1,480 fold in the MDR cell lines. In the sensitive cell line, LCC also dramatically decreased the IC₅₀ value of vincristine in the tested cell lines. Among the five active components, schizandrin A was found to be most potent (when tested at 25 and 50 µg/ml): the CE was 1,624 fold for vincristine, 32 fold for doxorubicin, and 113 fold for paclitaxel when used in combination. The mechanism for both LCC and schizandrin A was most likely related to induction of apoptosis and down regulate Pgp expression. In addition, Protein kinase C

expression was inhibited, further supporting inhibition of the MDR phenotype (Huang et al. 2008).

The in vivo efficacy of LCC was also determined in nude mice xenograft model implanted with KBv200 cells. The co-administration of LCC 300 mg/kg + vincristine 0.04 mg/kg significantly retarded tumor growth as compared to vincristine alone. However in this study, the LCC alone was not included as a control to allow differentiation of synergistic versus additive effect.

Green tea Green tea has been reported to have multiple pharmacological activities including inhibition of carcinogenesis, chemoprevention and antioxidant. The catechins ((–)-epigallocatechin gallate, EGCG), flavonoid (cyanidin, delphinidin), methyl-xanthines (caffeine, theobromine), and the vitamin group are the main active components (Sano et al. 1995). In addition, theanine, a major and specific amino acid in green tea (1–2 % of the dry weight of tea leaves) has been found to have antitumor effect (Sano et al. 1995). Theanine has been shown to inhibit doxorubicin efflux from Ehrlich ascites carcinoma cells, indicating a potential for CE. Its CE was further studied in P388 (leukemia cell line) and P388/DOX (doxorubicin resistant subline) in a mice xenograft model. At a daily dose of 10 mg/kg theanine combination decreased the tumor weight by 68 % compared with doxorubicin alone (Sadzuka et al. 2000).

Malaytea scurfpea fruit Liu et al. reported that the extract of this fruit increased the sensitivity to adriamycin in MCF-7/ADM cells at non-cytotoxic concentrations. The accumulation of rhodamine 123 and Pgp expression were completely abolished when MCF-7/ADM cells were treated with the Malaytea Scurfpea Fruit extract (Liu et al. 1997).

Rhei Rhizoma, Scutellariae Radix, Poria, Zingiberis Rhizoma, Zizyphi Fructus, Sophorae radix, and Asiasari Radix In a study using Hela cell line (not a resistant cell line) with Pgp expression, these herbs were found to enhance the cytotoxicity of paclitaxel at concentrations of 10 and 50 µg/ml. Among these 7 herbs, the Scutellariae Radix and Zizyphi Fructus were found to be relatively more toxic, with IC₅₀ values of 58 and 102 µg/ml respectively. The other five herbs at non-toxic concentrations were able to induce CE 1.7 to 5.0 fold. The rhodamine 123 uptake studies showed these herbal extracts did not induce any significant

uptake at the herbal concentrations of 10 and 50 µg/ml. Thus other mechanisms may be involved (Takara et al. 2005).

Rosemarinus officinalis Labiatae Rosemarinus officinalis Labiatae or Rosemary is a commonly used herb with potent chemopreventive activity in vivo. The CE of Rosemary was investigated using R65, an MCF-7-derived cell line which expresses Pgp and has acquired resistance to doxorubicin and vinblastine, and MCF clone 10.9 (a stable MDR1-transfected MCF-7 cell line). The results showed that Rosemary extract (16.5 µg/ml) significantly enhanced the inhibition of cell growth by vinblastine and decreased its IC₅₀. The IC₅₀ of Rosemary itself in both cell lines were around 30 µg/ml. The mechanism of the CE appeared to be related to the increase of intracellular accumulation of both doxorubicin and vinblastine via Pgp inhibition (Ciolino et al. 1999).

Tripterygium wilfordii We have studied the CE of various Tripterygium wilfordii (TW) extracts which were found to be capable of overcoming prostate cancer cell resistant to docetaxel in a dose-dependent manner at “non-cytotoxic” herbal concentrations (as evidenced by reducing IC₅₀ of docetaxel). These herbal extracts were also found to possess CE in combination with daunorubicin in K562(DOX) cells, a K562 resistant cell line to daunorubicin (Chow et al. 2014). The in vitro CE was verified by in vivo studies in a severely compromised immunodeficient (SCID) mice xenograft model implanted with PC-3/TxR (a docetaxel resistant prostate cancer cell line). The co-administration of an alcohol extract of TW (500 mg/kg daily) and docetaxel (20 mg/kg, weekly) was found to significantly inhibit tumor growth as compared to docetaxel alone. Neither docetaxel nor TW alone significantly inhibited tumor growth. These results also indicated that the CE from the herb can be separated from its cytotoxic activity which is an important consideration for product development (Chow et al. 2014).

Complex formula

Shenghe powder Shenghe Powder (SHP), consisting of Radix codonopsis pilosulae, Radix pseudostellariae, Radix scrophulariae, Rhizoma atractylodis macrocephalae, and additional 6 herbs, is a traditional Chinese medicine formula which has been shown to

induce tumor cell apoptosis, inhibit the proliferation of C6 (rat glioma cell line), decrease malignancy, and improve the survival time of the patients with postoperative high-grade glioma (Xia et al. 2011). Its ability to reverse the multidrug resistance was tested using the human SGC-7901 gastric carcinoma cell line and vincristine-resistant cell line (SGC-7901/vincristine). The intracellular concentration of vincristine was increased by SHP 10 $\mu\text{g/ml}$ and the corresponding expression level of Pgp and Bcl-2 was also downregulated. Thus, the SHP may be an effective herbal formula to overcome chemoresistance. Further studies however are needed to confirm this effect (Wang et al. 2007).

Shen-qi-jian-wei Tang Shu and Liu reported this Chinese medicine complex formula (composed of Astragalus, Semen Coicis, codonopsis, Atractylodes and 7 other substances) when combined with chemotherapeutic agents significantly inhibited the expression of MDR1 and LRP in the peripheral blood of gastric cancer patients (Shu and Liu 2008). Other clinical markers for the gastric cancer including CEA level and Karnofsky score were also improved in patients receiving the combination. The authors however did not specify whether these patients were chemoresistant patients and thus the efficacy of this complex formula for chemoresistant patients is unclear.

Si-Wu-Tang Si-Wu-Tang (composed of Radix paeoniae alba, Rhizoma chuanxiong, Radix angelicae sinensis and Radix rehmanniae preparata), has been reported to decrease adriamycin IC_{50} “index” in human erythrocyte leukemic cell line (K562/ADM) by increasing the intracellular accumulation of adriamycin. However Si-Wu-Tang did not affect the expression of Pgp (Shen et al. 2005). Further studies are needed to confirm these results.

Yi-qi-yang-ying-fang This formula (composed of Astragalus, atractylodes and Radix Glehniae) has been reported to sensitize the effect of cisplatin to human lung A549/DDP cell line (cisplatin resistant cell line) by reversing apoptosis to chemotherapy drug (Sun and Sun 2009).

Zuo Jin Wan Zuo Jin Wan (ZJW) consisting of the Rhizoma Coptidis and Fructus Evodiae (6: 1, w/w), is a traditional Chinese medicine formula. Rhizome Coptis chinensis Franch, the main component, has

previously been reported to have the MDR reversing effect (Min et al. 2006). The CE of ZJW in combination with oxaliplatin, diamminedichloroplatinum, or mitomycin in human colorectal cancer (HCT116), gastric cancer (SGC7901), and hepatic carcinoma (Bel7402) cell lines as well as their corresponding MDR sublines (HCT116/L-OHP, SGC7901/DDP, and Bel/Fu) was investigated. The ZJW alone induced only slight toxicity to all these three MDR cell lines with $\text{IC}_{50} > 80 \mu\text{g/ml}$. At the herbal concentration of 50 $\mu\text{g/ml}$ (which inhibited $<10\%$ cell viability) ZJW decreased the IC_{50} of all three drugs with CE values of 2.2, 1.8, and 1.4 for oxaliplatin, diamminedichloroplatinum, and mitomycin respectively. The chemosensitizing effect of ZJW at doses of 1,027.5, 2,055, and 4,110 mg/kg in combination with oxaliplatin (5 mg/kg) was further tested in vivo using a mice xenograft model implanted with HCT116/L-OHP. The combination significantly retarded tumor growth (in a dose dependent manner) as compared to oxaliplatin alone. ZJW alone did not show any anti-tumor effect. These results indicated that ZJW is capable of inducing a significant CE of oxaliplatin in the MDR HCT116 cells both in vitro and in vivo (Sui et al. 2013). The mechanism was related to a significant increase in intracellular oxaliplatin via downregulation of Pgp expression shown by the Western blot and MDR1-promoter luciferase assay. However its effect on Lung resistance protein (LRP) and MRP2 was not obvious (Sui et al. 2013).

Future perspective on the development of herbal medicine for CE

Based on the review of literature, the quality and quantity of research on the chemosensitizing herbs is inadequate, especially pertaining to in vivo and clinical effects relevant to product development. Up to now, the research work has been primarily focused on in vitro studies. In view of the exciting potential of herbal medicines that may be uniquely applicable in overcoming chemotherapy resistance, future research work should target the development of viable herbal products for therapeutic use. The use of appropriate in vitro screening, in vivo animal model and clinical studies are essential in generating relevant data for product development, thus key concepts to improve such studies are briefly discussed below.

In vitro screening of chemosensitizing herbs

Since many herbs are available world wide, screening of their CE is a first step in the discovery of potent chemosensitizing agents. A rapid quantitative method that would take into account not only the potency of CE but also safety of the herb is highly desirable, for comparing different herbs or extract during the screening phase.

The conventional method of assessing cytotoxic effect by incubating resistant cancer cells with herb-drug combination followed by reading of cell viability in the presence of sulforhodamine B is very time consuming and inefficient. An imaged based approach by using fluorescent or luminescent dye for assessing cell viability should be more advantageous (Worzella et al. 2013).

In quantifying the CE, the use of the combination index, although commonly employed for drug–drug interaction studies (Chou 2006), may not be the most suitable measure for the CE. We believe the use of a chemosensitizing utility index (CUI) that combines potency with “safety” (Wang et al. 2011) can yield useful information in comparing different herbs or different extracts. CE and CUI can be calculated as: $CE = IC_{50D}/IC_{50H+D}$; $CUI = CE \times (IC_{50H}/ConH)$, whereas D is the drug, H + D is the specific combination and ConH is the herb concentration in the combination (Wang et al. 2012).

Another useful concept that could be incorporated in in vitro studies is the determination of “safety margin” (SM). SM is the ratio of minimum toxic concentration (MTC) to the effective concentration of the drug (Arienti et al. 2008). For herb-drug combination, SM can be calculated as: $SM = MTC/C_{ce}$ where C_{ce} is the concentration of chemosensitizing effect, i.e. IC_{50} of the anticancer drug when in combination with the herb (IC_{50H+D}) that achieved significant CE (where H is the herb and D is the anticancer drug concentration). The MTC is usually determined from exposure of the normal peripheral blood cells to the herb-drug combination (e.g. set at IC_{20}).

We believe CUI or SM can be a useful tool to compare various chemosensitizing agents. The one that produced best CUI or SM with sufficient CE can be considered as the most suitable candidate for further in vivo testing. We have utilized this approach and found that our ethanol extract of *Tripterygium*

wilfordii yielded the best CE and CUI values compared to other extracts and was later verified by a positive animal xenograft study (Wang et al. 2011; Chow et al. 2014).

Animal model

The in vivo animal study is an essential step in validating the activity of the chemosensitizing herb before clinical study. Although several in vivo animal cancer models including carcinogen-induced spontaneous model, genetically engineered mouse model (GEM), and xenograft model (Richmond and Su 2008) have been commonly reported, these models need to be adapted for studying chemoresistance with the use of resistant human cancer cells. Thus the relevance of the carcinogen induced spontaneous model and GEM model for investigating chemoresistance is questionable. For the GEM model, chemoresistance is always restricted to single gene mutation, which cannot represent multiple drug resistance. In addition, these two models are actually mice tumor model and not human tumors. The xenograft model with subcutaneous tumor induced by human resistant cancer cells in immunocompromised mice (such as athymic nude mice, severely compromised immunodeficient (SCID) mice) can provide a simple proof of concept. However, the behavior of tumor cells grown subcutaneously can be significantly different when compared to those implanted into either the tissue of origin or implanted into an organ (Kim 2009). In addition, tumor cells implanted directly into their tissue of origin tend to grow better due to similarity of the microenvironment. Therefore, orthotopic models are more clinically relevant and better predictors of drug efficacy (Fleming and Brekken 2003).

One of the difficulties in designing an animal model to assess the chemosensitizing activity of the herb is the selection of the right dose for use in the herb-drug combination. In order to choose the right dose, both the efficacy and safety issues must be taken into account. The dose of the chemotherapeutic drug can be determined by the pharmacokinetic study with a target dose to achieve the minimum effective plasma/blood concentration (MEC) but lower than the minimum toxic concentration (MTC). Since determining the pharmacokinetics of the herb is not possible due to multiple components, a maximum tolerated dose (MTD) should be determined prior to selection of

the dose for the chemosensitizing activity study. In carrying out animal studies, a wide variety of study design has been reported. We believe a minimum of 4 groups comprising negative control, anti-cancer drug, herb, as well as the drug combination group are needed. Multiple doses of herb or drug groups as well as the use of xenograft induced by sensitive cancer cells should preferably be included. Also in planning the animal study, the tumor samples should be collected for biological analysis and gene expression studies for determination of the molecular mechanism.

Clinical studies

In developing any drug product, including herb products for CE clinical studies for safety (phase 1) and efficacy (phase 2 and 3) are required after *in vitro* and animal studies. Because many herbal products have already been used in human subjects, the safety associated with certain dose (e.g. maximum tolerated dose) of such herbs may be already known and the traditional phase 1 studies may not be required. For the efficacy study (phase 2 and 3), the endpoints (desired clinical outcomes) from the chemosensitizing herb should be the improvement of both clinical remission of cancer and long term survival. However, achievement of clinical remission alone while desirable, may not lead to long term survival benefit, if resistant CSC are present and not eliminated. Thus in planning clinical studies, the target for mechanism of resistance relating to the clinical endpoint is important and should be identified before a clinical trial, e.g. randomized controlled (or comparison) trial. It would be most desirable to identify a relevant endpoint that can be targeted for prevention of resistance.

Another important issue for clinical study is the quality of the herbal product. Since there is no uniformly recognized control (QC) measure for herbal medicines at the present time, the use of active chemical markers plus the *in vitro* CE value (representing biological effect of the herb) could serve as reasonable QC markers for the chemosensitizing herb. However, further work on specificity and sensitivity and relevance of such markers for QC purposes should be established for each specific herbal product.

In designing and carrying out a clinical trial, the selection of suitable patient to demonstrate CE is a challenging step. A case in point is the chronic

myelogenous leukemia in which the clinical remission is usually judged by hematologic (peripheral blood counts, the differential and spleen size), cytogenetic (%Ph-positive metaphases) and molecular changes (BCR-ABL levels). Remission may last 3 years and primary resistance is uncommon in patient who started imatinib in the chronic phase. Thus, any study seeking to determine the effect of a sensitizing agent to overcome resistance in such patients will take a long time. Also using the randomized control design to determine the efficacy of prevention of chemoresistance, a large patient population may be required, unless markers of development of specific drug resistance can be identified. Thus research leading to identification of specific marker or mechanisms of resistance for different drugs/types of cancer is urgently needed to facilitate future development of products to overcome cancer chemotherapy resistance.

Mechanism of resistance

Demonstration of inhibition of specific mechanisms of chemoresistance by the chemosensitizing herb is useful in validating the activity of the herbal product. Although determining the molecular mechanisms from the *in vitro* cell line is relatively simple, determining these mechanisms *in vivo* and in patients is more relevant but can be difficult. When tumor cells grow *in vivo*, it is not easy to precisely measure the mRNA and protein level since these may vary depending on the area dissected. In order to get more reliable results, multiple samples from the same tumor should be collected rather than one sample from one tumor. Since the dynamic process of the gene expression is difficult to determine, the selection of sampling time and treatment duration may alter the results obtained and need to be considered in interpretation of the results.

Recently, the cancer stem cell (CSC) was found to be able to confer resistance. The hedgehog (Hh), Notch, Wnt pathways have been implicated in CSC self-renewal, survival and drug resistance (Crea et al. 2011; Domingo-Domenech et al. 2012; Merchant and Matsui 2010). Therefore, the inhibition of these pathways may be related to CE and can provide an important mechanistic approach for future investigation of chemoresistance and prevention.

Summary and conclusion

In view of the magnitude of problem in chemoresistance and lack of progress with conventional agents, development of herbal products to overcome this therapeutic problem may offer advantage by (1) targeting multi-mechanisms of resistance, (2) availability of large number of herbs from world wide resources for screening, and (3) reducing the need for phase I (safety) study, since many herbs have already been used in human subjects. Despite these advantages and the exciting potential of the herbal medicines as chemosensitizing agents, the quality and quantity of research toward product development is inadequate at present based on our review of the literature. Future work aiming at proper in vitro screening, use of appropriate animal model and conducting relevant clinical studies are needed to enhance the success of herbal product development for the treatment of chemoresistance. In this regard, collaboration among different disciplines are strongly encouraged, especially among the pharmaceutical scientists, phytochemists, molecular bio scientists and clinicians, to work in a concerted manner to further investigate and develop many available herbs as chemosensitizing agents to overcome chemotherapy resistance in the future.

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